ResolveAI: Knowledge Transfer for New Engineers

Oct 31, 2026 · @Ratnesh

1. Start here

ResolveAI helps an on-call engineer investigate an incident. It gathers evidence from read-only systems, proposes root causes with citations, and drafts a root cause analysis (RCA) for a person to verify. It never changes anything in the systems it inspects.

This document describes branch `feature/dotnet-port`, commit `a247662`. That branch is not merged to `main`. For design rationale and the future plan, see `docs/ResolveAI-Architecture-and-Design.pdf`.

Five facts to remember

1. **Code gathers the evidence; the model only reasons over it.** Stages that collect logs, deployments and past incidents use no AI.
2. **Every piece of evidence has a typed ID**, such as `log:…` or `mcp:3`. The server removes any citation it cannot match to a real record.
3. **Connections to monitored systems are read-only.** Each system sits behind its own MCP server with fixed tools. No tool runs arbitrary SQL or shell commands.
4. **A person decides.** An incident only becomes verified after someone records the cause and reviews the RCA.
5. **The default mode uses no AI at all.** Recorded mode replays written analyses, so what you see in a demo does not prove real model quality.

Current state

Area	State
Blazor Server UI and HTTP API	Working
Investigation pipeline in recorded mode	Working
Four MCP servers with 17 tools	Working, including real SQL Server checks
RCA drafting, editing, versions, Markdown export	Working

Area	State
Automated tests	205 cases at last run: 196 unit and API, 9 browser
Docker image	Built and tested locally
Live Claude	**Never run against the real API.** Tested only against a fake API
GitHub CI	Workflow written, not yet run on GitHub
User accounts and roles	Not built; a shared access code gates paid runs
Data	Synthetic, except optional SQL Server checks

Suggested first week

Day	Do this	Sections
1	Run the app and the tests. Click through every page.	2, 3
2	Read how the parts fit. Follow one investigation through the code.	4, 5
3	Read the key components, the data model and the configuration.	6, 7, 8
4	Make a practice change: add a tool to a scratch branch, using the task guide.	9, 10
5	Reset demo data, break something on purpose and diagnose it. Read the security rules and the known traps.	11, 12, 13

Section 14 ends with a checklist to confirm the handover is complete.

2. Set up and run it

You need only the .NET 10 SDK to run the app. It starts on `http://127.0.0.1:8787`, seeds an empty database on first start, and needs no API key.

Prerequisites

Tool	Needed for	Required?
.NET 10 SDK	Building, running and testing	Yes
Git	Source, and the deploys server	Yes
Docker	Container run and Linux tests	Optional
SQL Server (local)	Real database checks and `MSSQL_TEST=1` tests	Optional
Node	Only to regenerate `wwwroot/app.css`	Optional
Anthropic API key	Live mode. Each run costs money, so ask before using one	Optional

First run

```
git clone <repository url>
cd resolveai
git checkout feature/dotnet-port
dotnet run --project src/ResolveAI.App
```

Open `http://localhost:8787`, pick **INC-1042**, and select **Run investigation**.

Confirm it works

```
curl http://localhost:8787/api/config      # mode should be recorded
curl http://localhost:8787/api/systems     # systems should show status
connected
curl "http://localhost:8787/api/incidents" # should list INC-1042
```

The database system shows as off until you set up SQL Server. That is expected (section 8).

Run in Docker

```
cp .env.example .env          # optional, for live runs
docker compose up -d --build
docker compose ps             # wait for "healthy"
```

The port is published on `127.0.0.1` only. Data lives in the named volume `resolveai-data` and survives rebuilds.

Run the tests

```
dotnet test tests/ResolveAI.Tests                # no external services needed
dotnet test tests/ResolveAI.E2E                  # real browser, temporary
database, free port
MSSQL_TEST=1 dotnet test tests/ResolveAI.Tests   # also uses local SQL Server
docker build --target test .                     # same unit tests on Linux
```

Browser tests need Chromium installed once: `pwsh tests/ResolveAI.E2E/bin/Debug/net10.0/playwright.ps1 install`.

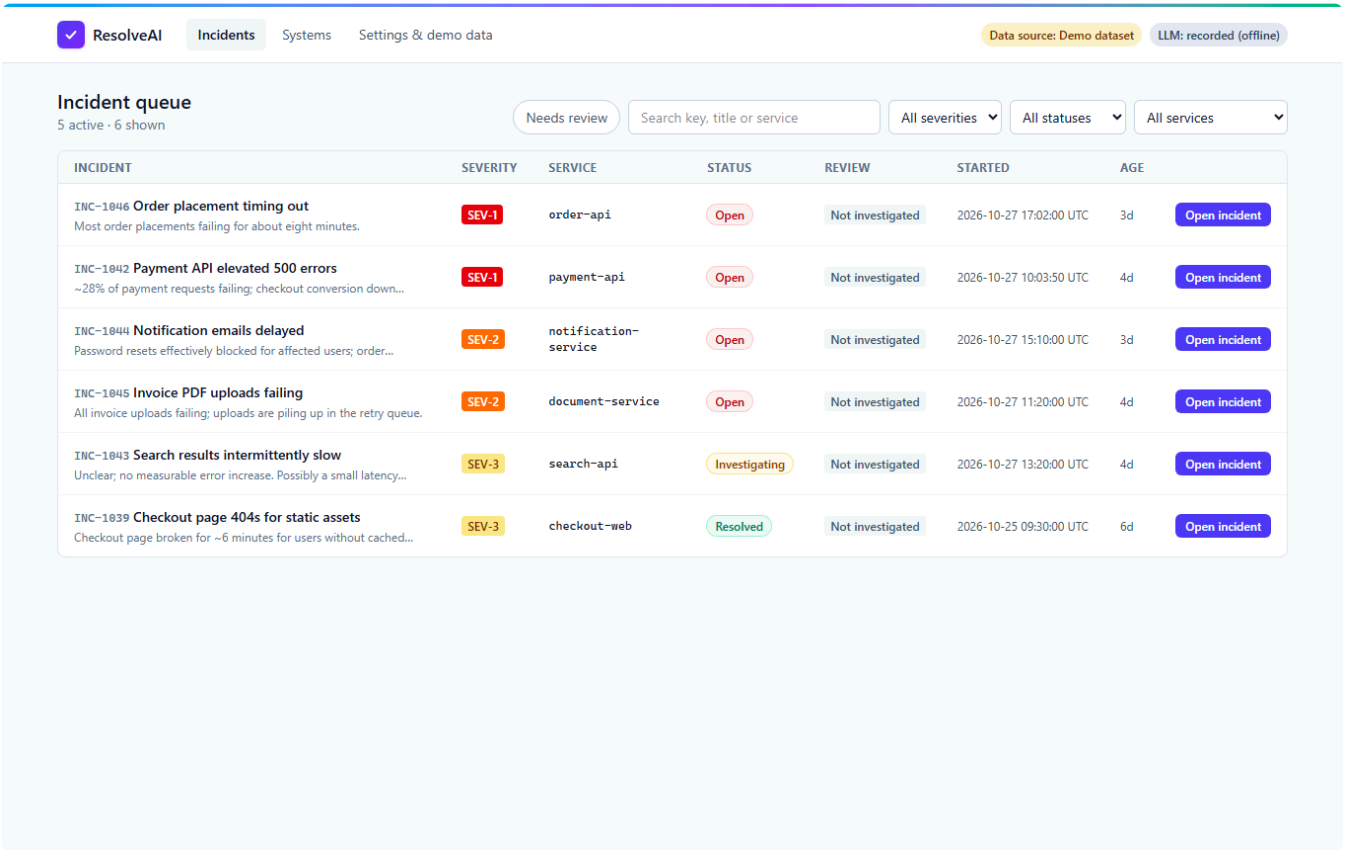
If the app is already running, stop it before building. A running process can lock the build output on Windows.

3. Tour of the app

The app has three pages in its top menu, plus an incident page with seven tabs. These screenshots come from recorded mode on demo data, using INC-1042 as the example.

Two badges at the top right appear on every page. They show the data source and whether a model is being called. Check them first when something looks odd.

Incident queue (`/`)

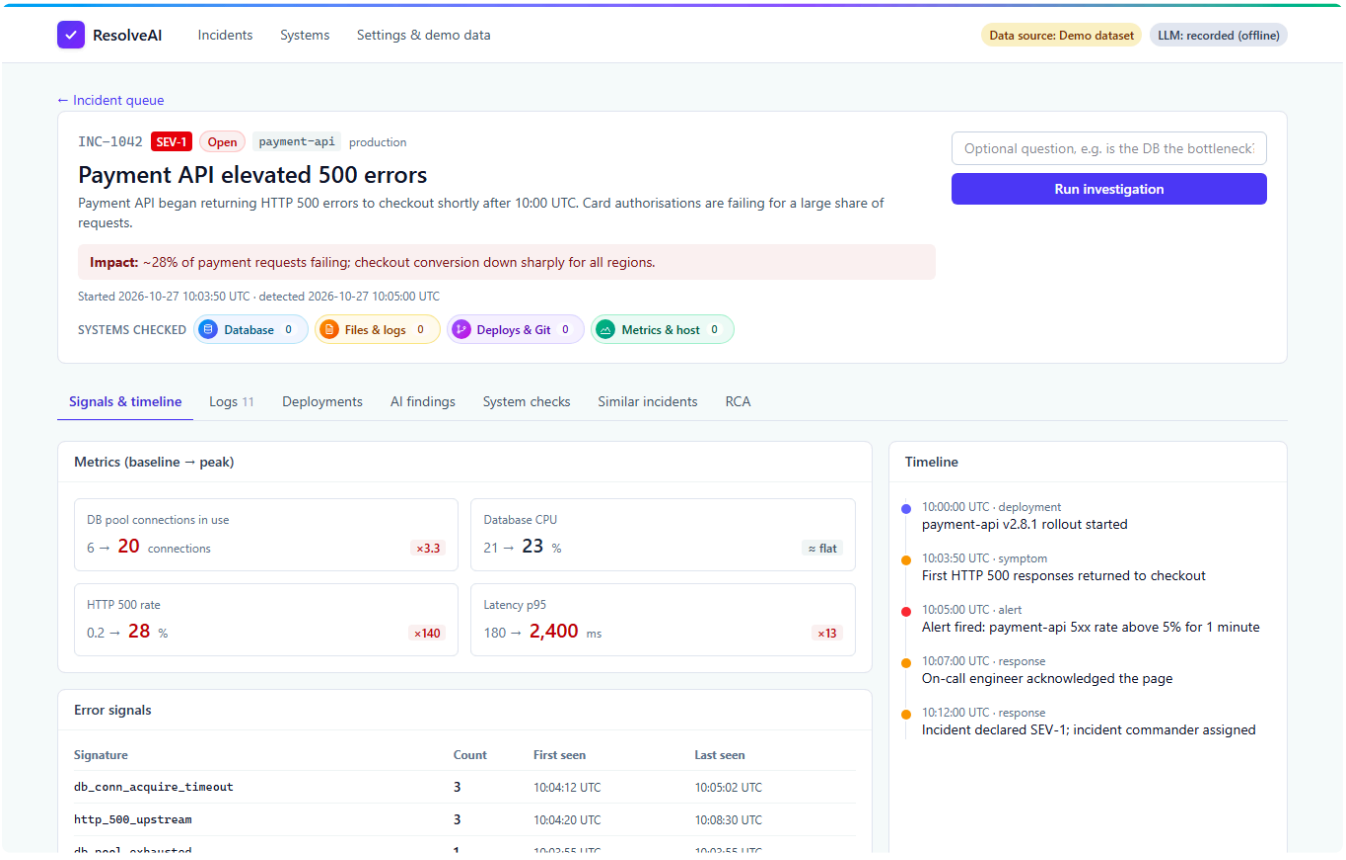


This is the home page. Filter by severity, status or service, search by key or title, or use **Needs review** to see incidents waiting on a person. The Review column has three values: Not investigated, Needs review and Verified.

The six demo incidents teach different lessons, so try more than one:

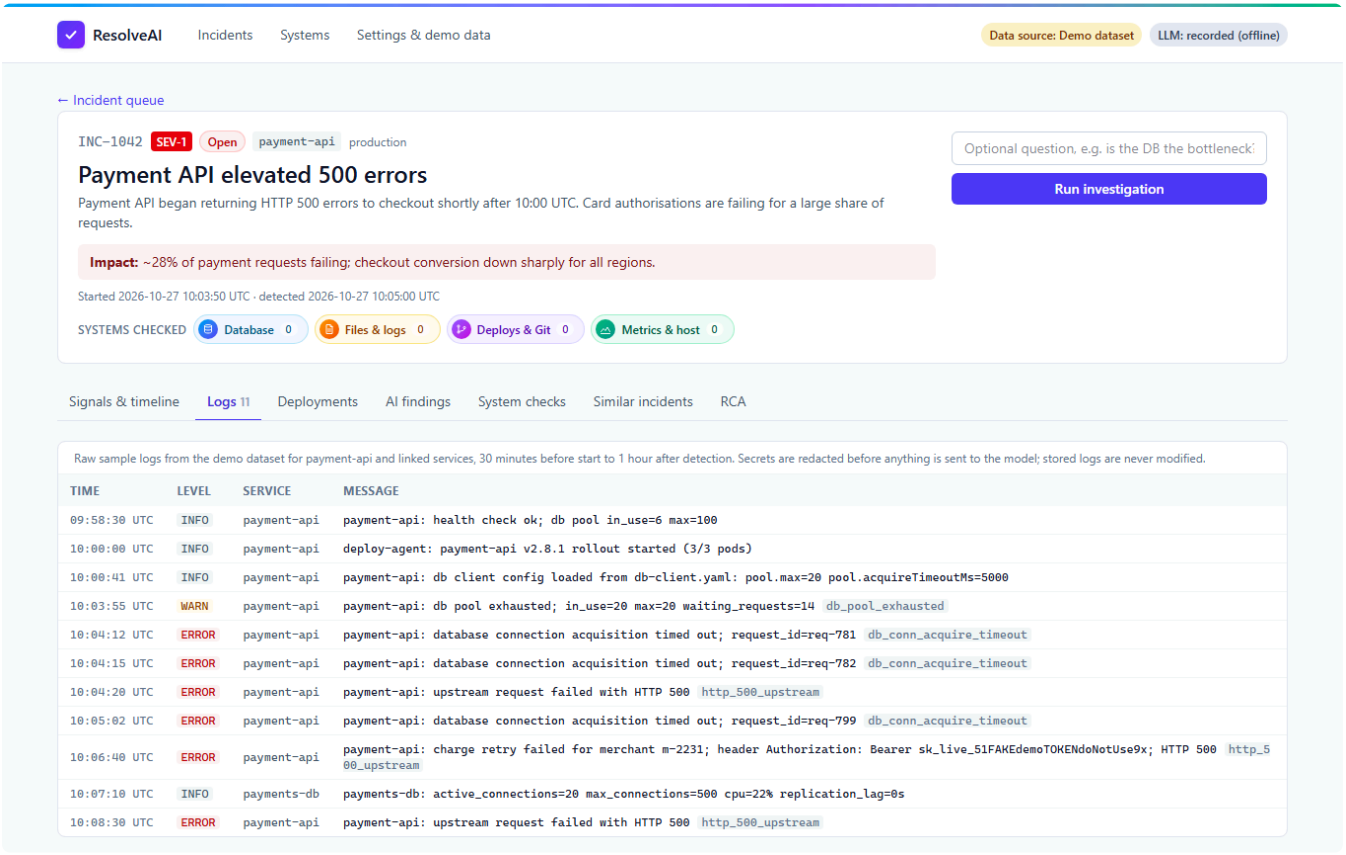
Incident	Service	Analysis used in recorded mode
INC-1042	payment-api	Written analysis: connection pool regression after a release
INC-1045	document-service	Written analysis that adapts to the checks that ran
INC-1046	order-api	Written analysis; uses real SQL Server blocking when set up
INC-1043	search-api	Rule-based
INC-1044	notification-service	Rule-based
INC-1039	checkout-web	Rule-based; already resolved

Incident page (`/incidents/{id}`)



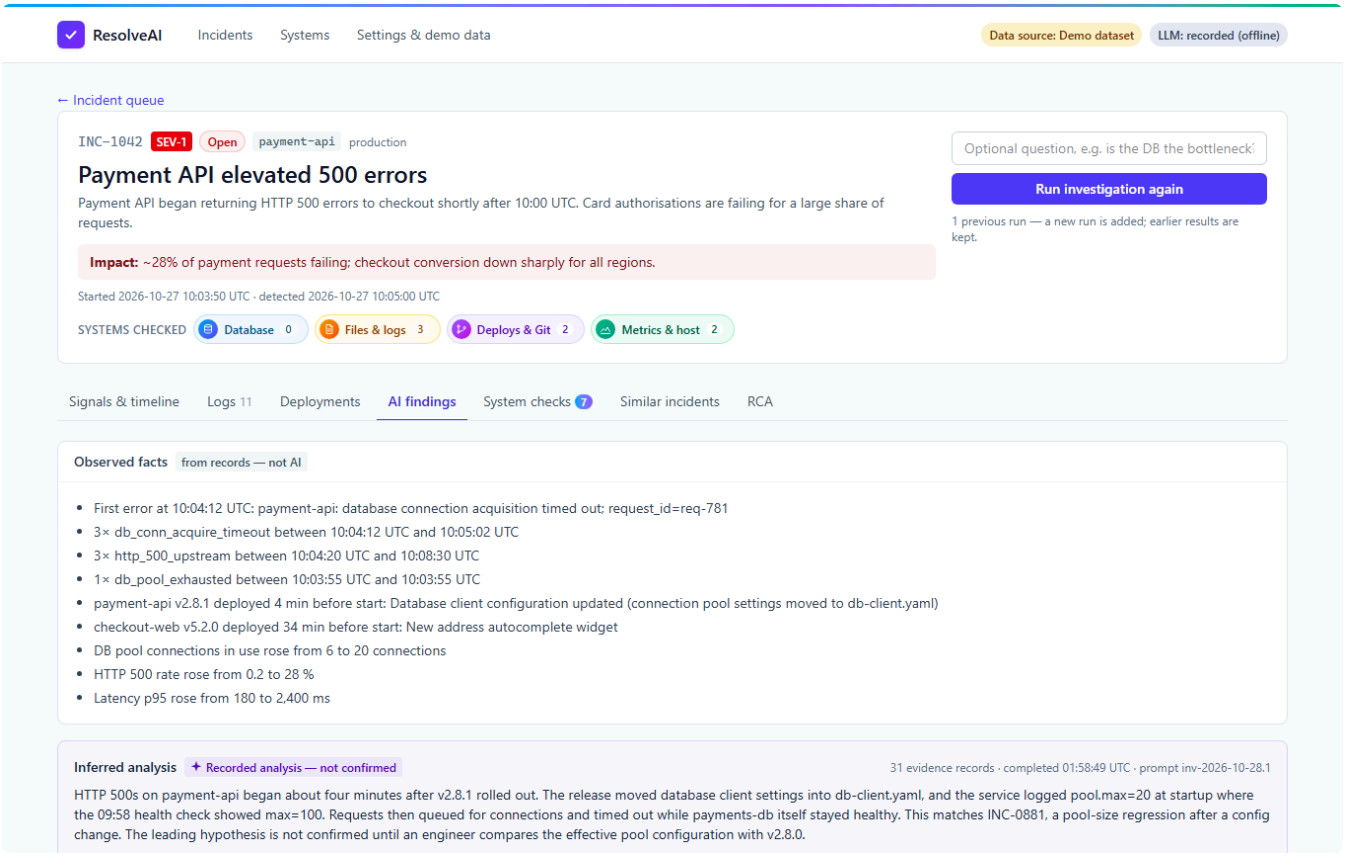
The header gives the summary and impact, and holds **Run investigation** with an optional question. The **Systems checked** strip counts the checks made against each system. The first tab shows metrics, the timeline and error signals, all taken from records with no AI involved.

Logs tab



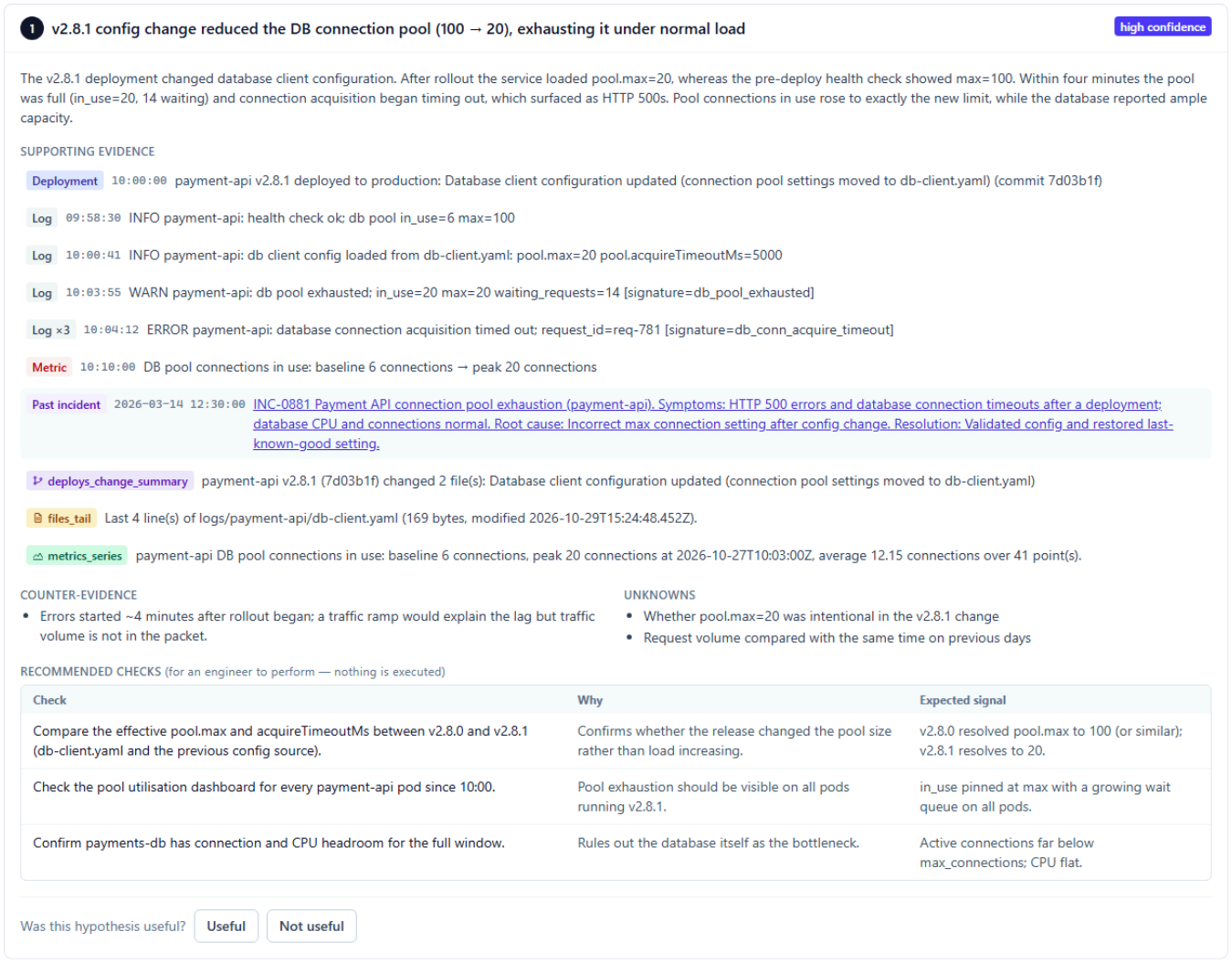
This shows raw logs from 30 minutes before the start to one hour after detection. Line 10:06:40 contains a deliberately fake token, `sk_live_51FAKE…`. You see it here because stored data is never changed. It is redacted only in the copy sent to the model, which is a useful thing to verify yourself.

AI findings tab



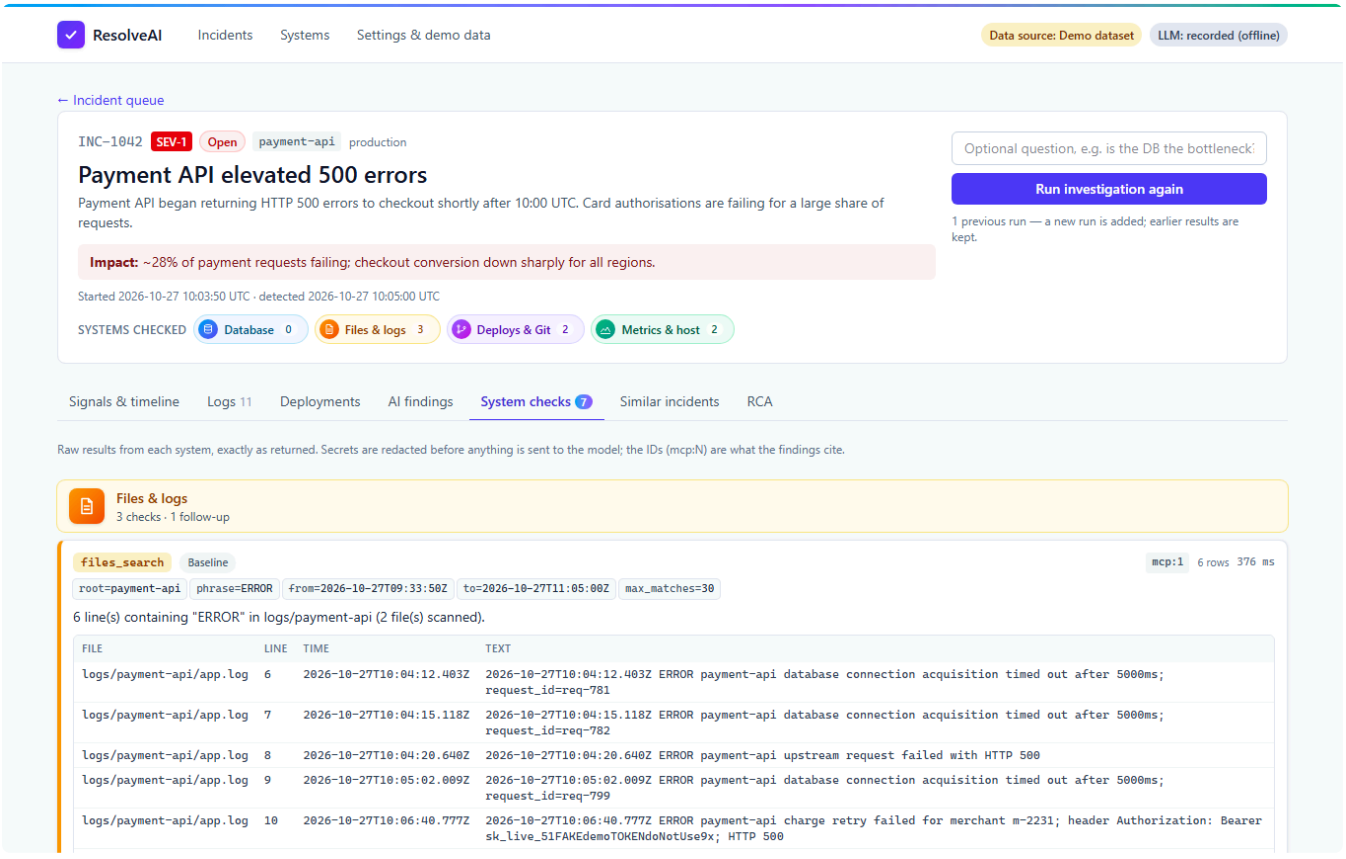
This tab fills in after a run. Its layout matters: **Observed facts** come from records, and **Inferred analysis** comes from the provider. The label here reads *Recorded analysis*, so no model produced it. Below this sits **How this was produced**, a trace of each stage with timings and counts.

Each hypothesis is laid out as a case to check:



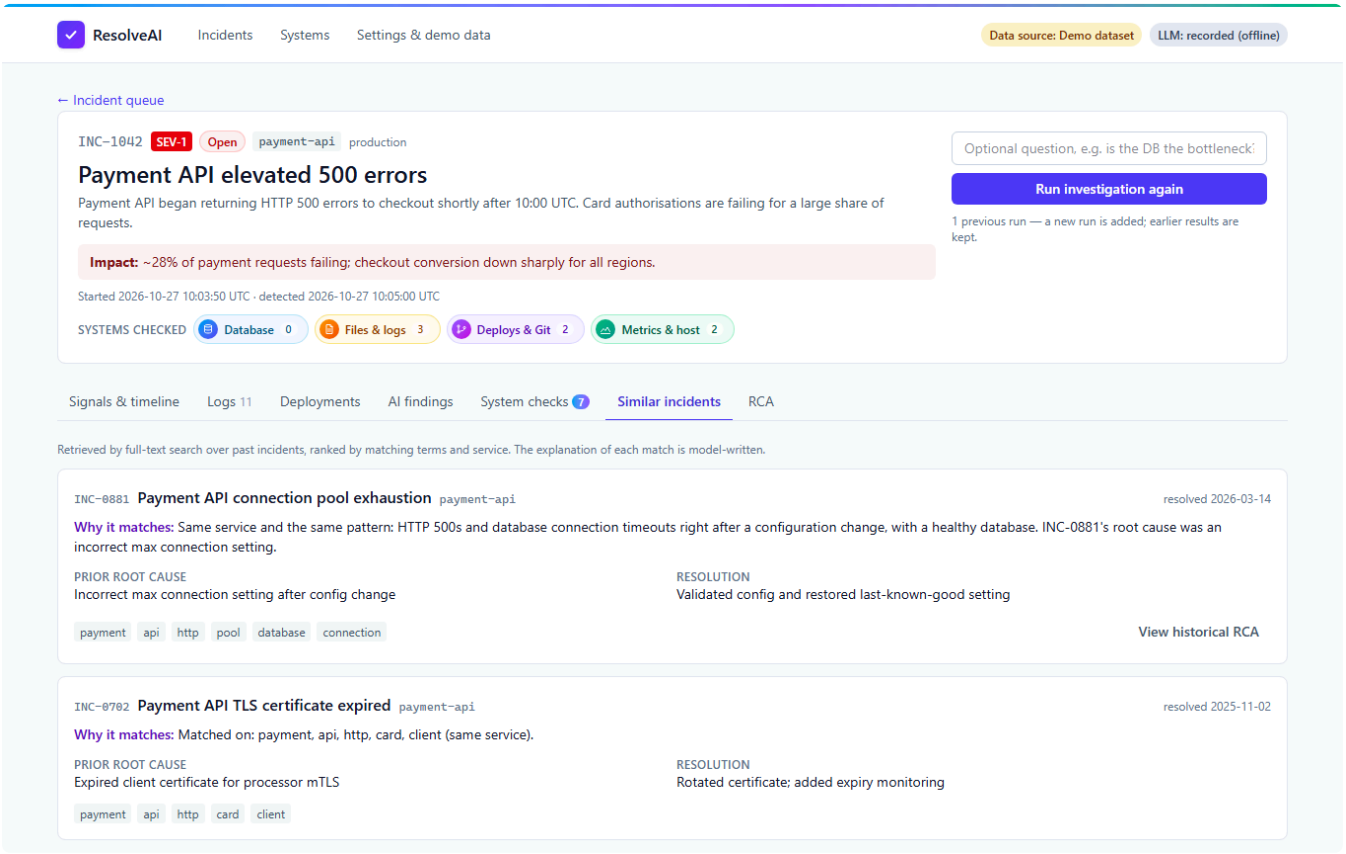
Select an evidence row to jump to its source record. Recommended checks are for a person to perform, and nothing is executed.

System checks tab



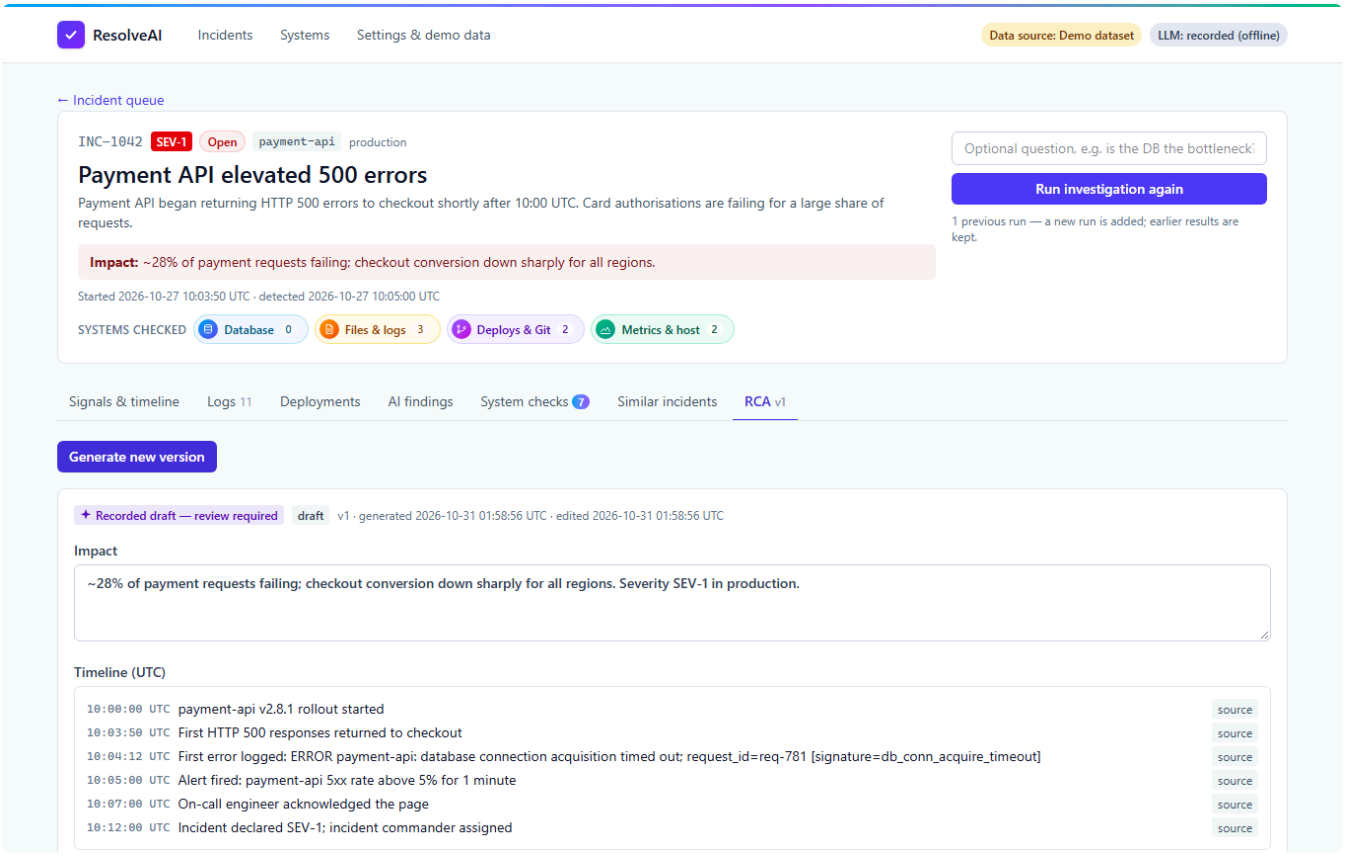
This shows the raw result of every MCP call, grouped by system. Each has a label of **Baseline** or **Follow-up**, its arguments, and its `mcp:N` ID. Those IDs are what the findings cite. Use this tab first when you doubt a finding.

Similar incidents tab



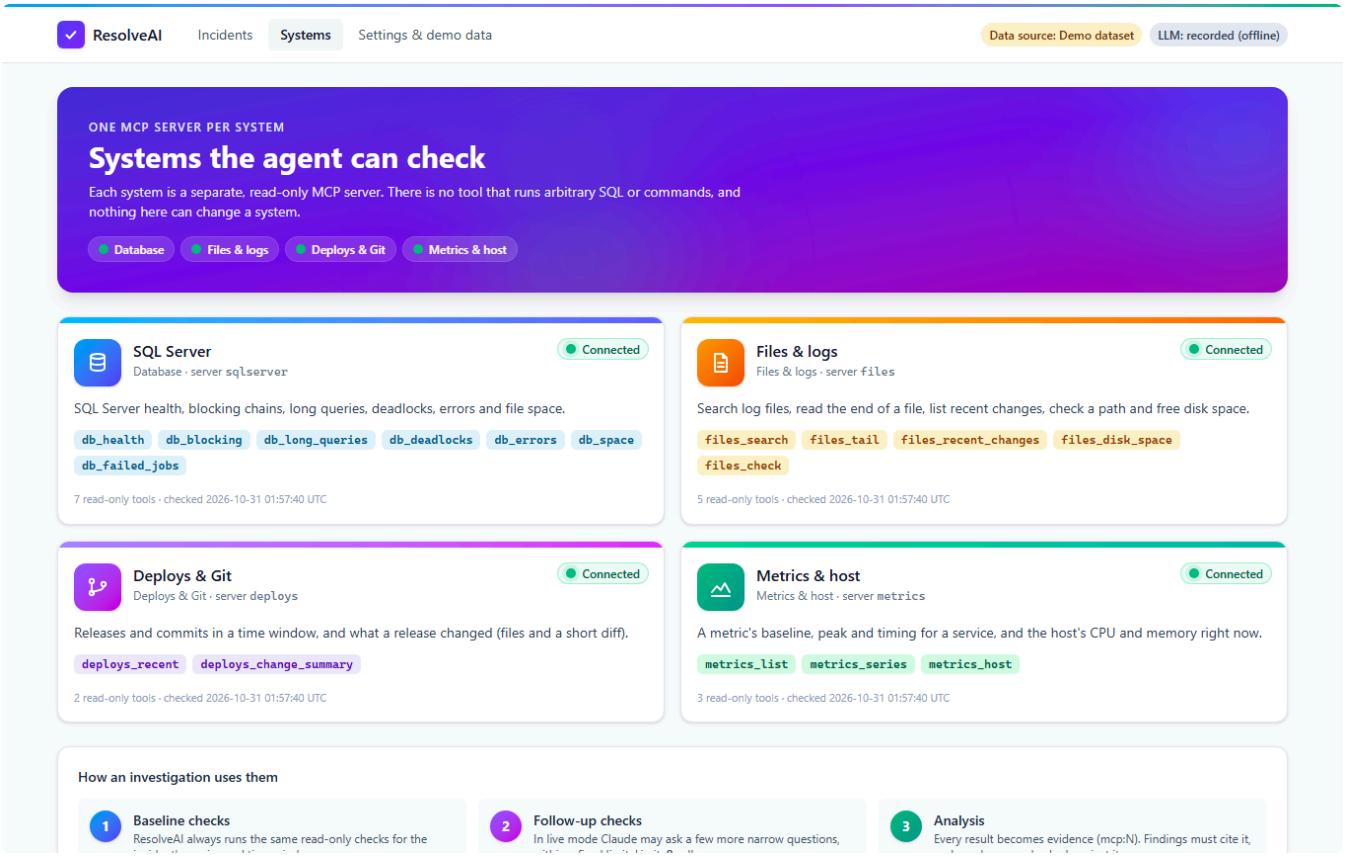
Code chooses these using full-text search. The model can only write the *Why it matches* text, so it cannot invent a past incident.

RCA tab



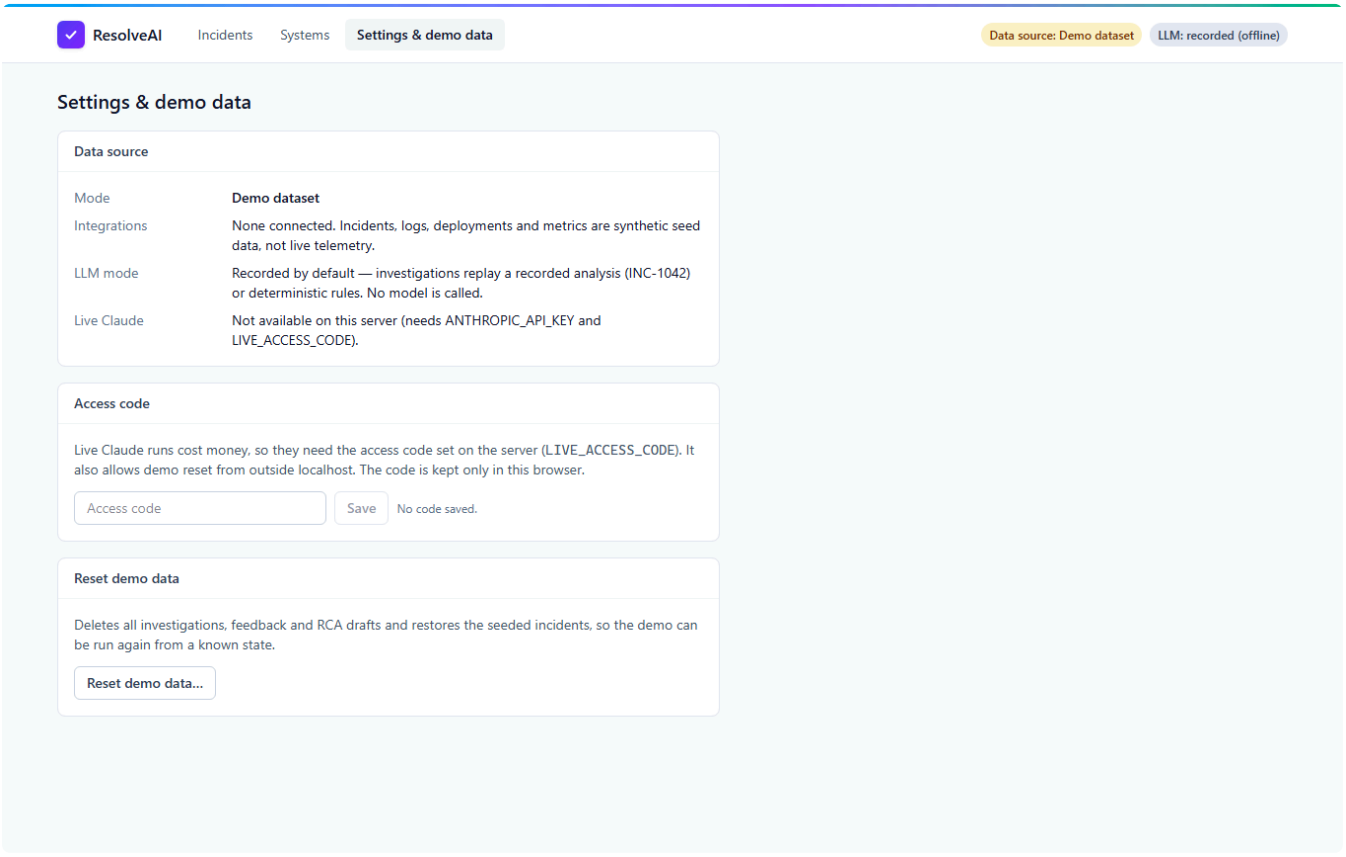
Generate a draft after a run completes, then edit it and mark it reviewed. Generating again creates a new version, so it never overwrites someone's edits. Export it as Markdown by copying or downloading.

Systems page (`/systems`)



This shows each MCP server's status and tools. On a fresh clone SQL Server shows as off; this machine has it turned on in its local `mcp.config.json` .

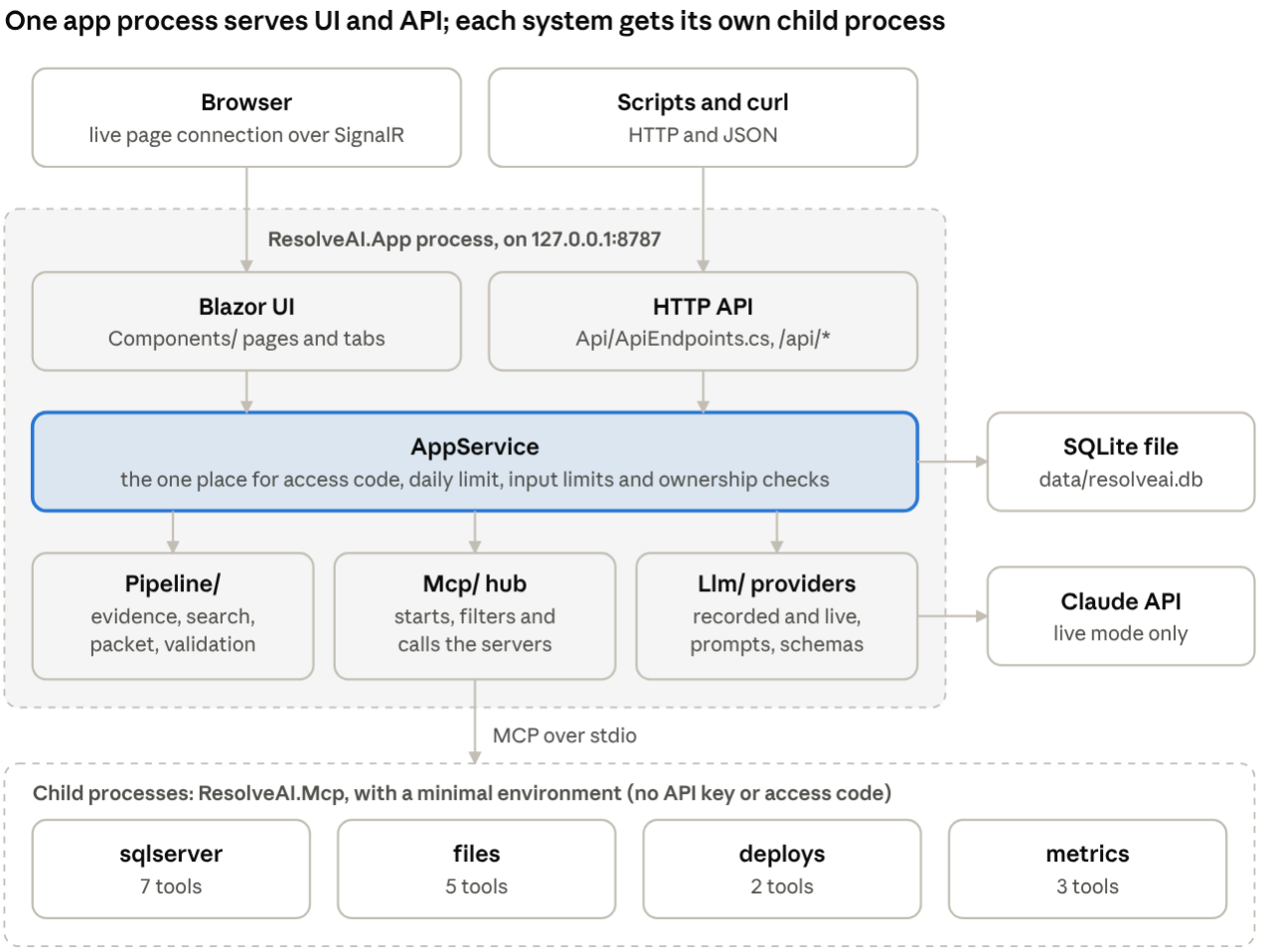
Settings page (`/settings`)



This holds the access code for live runs and **Reset demo data**, which deletes all investigations, feedback and RCA drafts. Its text still names only INC-1042 as a recorded analysis; three incidents have one.

4. How the system fits together

The UI and the API are two doors into the same `AppService`, so a rule added there applies to both. Everything runs on one machine: one web process and four child processes.



runtime architecture · one app process, four MCP child processes

The UI calls `AppService` directly instead of going over HTTP. The API exists for scripts and the RCA export. The child processes start when the app boots, and a system that fails to start never stops the app.

Three projects

Project	Type	Holds
`src/ResolveAI.Contracts`	Library	Shared DTOs, JSON rules, evidence IDs
`src/ResolveAI.Mcp`	Executable	The four MCP servers, chosen by argument: `ResolveAI.Mcp files`
`src/ResolveAI.App`	Web app	UI, API, pipeline, providers, data access

Where to find things

If you need to change	Look in
App startup and middleware	`src/ResolveAI.App/Program.cs`
Settings and environment variables	`AppSettings.cs`
How services are wired together	`AppDeps.cs`
HTTP routes and error mapping	`Api/ApiEndpoints.cs`
Access, limits and ownership rules	`Services/AppService.cs`
The investigation stages	`Services/Investigations.cs`
RCA generation and editing	`Services/Rca.cs`
Queue listing and filters	`Services/Incidents.cs`
Evidence gathering	`Pipeline/EvidenceLoader.cs`
Similar-incident search and scoring	`Pipeline/Similar.cs`
Packet building and bounds	`Pipeline/Packet.cs`
Redaction rules	`Pipeline/Sanitizer.cs`
Checks on model output	`Pipeline/Validate.cs`
Prompts and output schemas	`Llm/Prompts.cs`, `Llm/Schemas.cs`
Recorded analyses	`Llm/RecordedProvider.cs`
Real Claude calls	`Llm/ClaudeProvider.cs`
Which checks every run performs	`Mcp/Baseline.cs`
Starting and calling MCP servers	`Mcp/Hub.cs`
A tool's behaviour	`src/ResolveAI.Mcp/*Server.cs`
Tables	`Data/Migrations/*.sql`
Demo incidents	`Data/Seed.cs`
Pages and tabs	`Components/Pages`, `Components/Incident`
Styles	`Styles/app.css`, which generates `wwwroot/app.css`

Paths in the table are inside `src/ResolveAI.App` unless they start with `src/`. Other top-level folders are `tests/`, `demo-systems/` (synthetic logs, deploy feed, metrics and SQL scripts), and `tools/css/`.

5. Follow one investigation through the code

A run returns HTTP 202 immediately, then continues on a background task through seven stages. Only stages 5 and 6 depend on the provider. The best way to learn this is to set a breakpoint in `Investigations.Run` and step through INC-1042 in recorded mode.

Before the run starts

`AppService.StartInvestigation` makes these checks in order:

1. The incident must exist, or the call returns 404.
2. The question must be 500 characters or fewer, and the mode must be `recorded` or `live`.
3. If this incident already has a queued or running investigation, its ID is returned and no second run starts.
4. A live run needs the access code and must be under the daily limit, or the call returns 400, 403 or 429.
5. `Investigations.Start` inserts a row with status `queued` and starts the background work.

The seven stages

These are all in `Investigations.Run`, in `Services/Investigations.cs`.

#	Stage	Code	AI?	If it fails
1	Gather logs, deployments, metrics, timeline, runbooks	`Pipeline/EvidenceLoader.cs`	No	Run fails
2	Search incident history	`Pipeline/Similar.cs`	No	Run fails
3	Baseline system checks	`Mcp/Baseline.cs`, `Mcp/Hub.cs`	No	A failed check is recorded; the run continues

#	Stage	Code	AI?	If it fails
4	Build the evidence packet	`Pipeline/Packet.cs`, `Pipeline/Sanitizer.cs`	No	Run fails
5	Follow-up checks	The provider, then the hub	Live only	Noted as an issue; analysis still runs. Limit 120 s
6	Analyse	`ILlmProvider.Investigate`	Live only	Malformed output retries once, then the run fails. Limit 180 s
7	Validate, score and rank	`Pipeline/Validate.cs`	No	Run fails

Results are then stored in one transaction with status `completed`, including hypotheses, evidence links, similar incidents and the trace. A failed run stores a safe, readable message.

What happens in the browser

The page polls the investigation every 1.5 seconds and shows the current stage text. On completion the tabs fill in. To follow it from the API:

```
curl -X POST http://localhost:8787/api/incidents/INC-1042/investigations \
  -H "Content-Type: application/json" -d '{}'
# returns an investigation id, with HTTP 202
curl http://localhost:8787/api/investigations/<id>
```

Worked example: INC-1042

1. **Evidence:** 11 log lines, including `pool.max=20` loaded straight after the v2.8.1 rollout, and a health check showing `max=100` before it.
2. **History:** INC-0881, a pool-size regression after a config change, ranks first because it matches on terms and service.
3. **Baseline checks:** `files_search` finds the errors, and `deploys_recent` and `metrics_series` find the release and the pool rising from 6 to 20.
4. **Packet:** the fake bearer token in the 10:06:40 log line is redacted here.
5. **Follow-ups:** recorded mode replays two calls: `deploys_change_summary` shows what changed, and `files_tail` reads `db-client.yaml`.

6. **Analysis:** the written analysis says v2.8.1 reduced the pool from 100 to 20.
7. **Validation:** each citation is matched against the packet. The result stays labelled as not confirmed.
8. **Person:** an engineer performs the recommended checks, records the verified cause, and generates the RCA.

Time limits to know

Operation	Limit
Analysis stage; RCA draft	180 s each
Follow-up stage	120 s
One Claude HTTP request	90 s, with 2 retries
MCP server start	20 s
One tool call, at the hub / inside the server	15 s / 10 s
Git command	8 s, with a 4 MB output cap
SQL connect / command	5 s / 8 s

If the app restarts during a run, startup marks that run as failed. There is no resume, so run it again.

6. Key components

Each component below does one job, and most can be understood by reading a single file of under 250 lines. `RecordedProvider.cs` and `Investigations.cs` are the two long ones.

Evidence loader

`Pipeline/EvidenceLoader.cs` is deterministic. For one incident it loads:

- **Logs:** the affected service or linked to the incident, from 30 minutes before the start to 1 hour after detection, up to 200 rows.
- **Deployments:** the same service in the previous 48 hours, and other services within 1 hour either side.
- **Other records:** metric snapshots, timeline events, and up to 5 runbooks or guides.

Similar incidents

`Pipeline/Similar.cs` builds search terms from the title, description, error signatures and change summaries, then runs an FTS5 query.

```
score = -bm25 x 10 + matched terms + (3 if same service)
keep  = at least 2 matched terms, at least 35% of the best score, top 5
```

Evidence packet

`Pipeline/Packet.cs` builds the only thing the model may see. Each item has a typed ID, and those IDs are the citation vocabulary.

Prefix	Source	Prefix	Source
`log:`	Log group	`doc:`	Runbook or guide
`dep:`	Deployment	`hist:`	Past incident
`met:`	Metric snapshot	`tl:`	Timeline event
`mcp:N`	System check result		

Log lines that differ only by IDs or numbers collapse into one group with a count. The packet is capped at 40 log groups and 40,000 characters. Excerpts stop at 600 characters, or 1,200 for tool results, and information logs are dropped first.

Redaction

`Pipeline/Sanitizer.cs` runs when the packet is built. It covers URL credentials, `Bearer` and `Basic` values, `sk_` / `pk_` / `rk_` keys, `sk-ant-`, `AKIA`, `gh*_` tokens, `password=`-style values, email addresses and card-like numbers.

Stored records are not changed, so the UI shows raw data. Redaction is pattern-based and a secret in an unusual format can pass through.

Validation

`Pipeline/Validate.cs` treats model output as untrusted.

Check	Behaviour
Citations	IDs not in the packet are removed. A hypothesis with none left is marked speculative and forced to low confidence

Check	Behaviour
Unsafe checks	Destructive suggestions such as `rm -rf`, `DROP TABLE` or `kubectl delete` are removed
Invented numbers	Numbers with a unit, or of 10 or more, that are not in the evidence are flagged, not deleted
Similar incidents	Explanations for anything outside the retrieved candidates are ignored
Bounds	5 hypotheses, 8 items per list, 5 checks
Ranking	`2 x confidence (1 to 3) + min(citations, 5) - counter-evidence`. Used for order only

LLM providers

Both implement `ILlmProvider`, so the rest of the pipeline does not know which is in use.

	`RecordedProvider`	`ClaudeProvider`
Network	None	Claude API, model `claude-opus-5-5`
Cost	Free	Paid; counts against the daily limit
Analysis	Written for 3 incidents, rule-based for others	Structured output, effort medium, 16,000 tokens
Follow-ups	Replayed from a fixed list	Claude chooses, effort low, 4,000 tokens, default limit 8
Proven by	Tests and demos	**Fake API tests only**

MCP hub

`Mcp/Hub.cs` starts each enabled server, keeps only tools marked `readOnlyHint`, routes calls, and restarts a crashed server once. `Mcp/Baseline.cs` plans the fixed checks for each run:

- `db_health`, `db_blocking`, `db_errors`, `db_deadlocks`, if the service has a database.
- `files_search` for `ERROR` and for `WARN`, in each log folder.
- `deploys_recent` for the service.
- `metrics_series` for each key metric.

MCP servers and tools

All four servers live in `src/ResolveAI.Mcp`. `ToolServer.cs` applies input checks, timeouts, read-only annotations and error results to every tool.

System	Tool	What it does
Database	`db_health`	Sessions, blocked requests, top waits, memory and CPU
Database	`db_blocking`	Blocking chains and the head blocker
Database	`db_long_queries`	Requests running at least n seconds
Database	`db_deadlocks`	Deadlocks from the system_health session
Database	`db_errors`	Errors from extended event sessions
Database	`db_space`	File sizes and log reuse for an allowed database
Database	`db_failed_jobs`	Failed SQL Agent job steps
Files	`files_search`	Plain-text search, optionally in a time window
Files	`files_tail`	Last lines of a text file
Files	`files_recent_changes`	Files modified since a time
Files	`files_disk_space`	Free and used space
Files	`files_check`	Exists, size, modified time, readable, writable
Deploys	`deploys_recent`	Feed deployments and git commits in a window
Deploys	`deploys_change_summary`	Files changed and a short patch
Metrics	`metrics_list`	Metric names for a service
Metrics	`metrics_series`	Baseline, peak and timing in a window
Metrics	`metrics_host`	Current CPU, memory and uptime

Every tool has these limits: 10 s timeout, 24-hour window, 200 rows, 64,000 characters of output and 500 characters per value.

To use the servers from Claude Code or Claude Desktop, copy `.mcp.json.example` to `.mcp.json`. That is a quick way to explore one tool in isolation.

7. Data and API

All application data is in one SQLite file: 14 tables, two full-text search tables and a `_migrations` table. Lists and structured values are stored as JSON text in columns.

Tables

Table	Holds	Notes
`incidents`	The incident record	Parent of most data
`log_events`	Raw logs	A trigger rejects any update
`deployments`	Releases	Optional link to an incident
`metric_snapshots`	Baseline and peak values	
`timeline_events`	What happened and when	
`knowledge_documents`	Runbooks and historical RCAs	Searched through `knowledge_fts`
`historical_incidents`	Past incidents	Searched through `historical_fts`
`investigations`	One row per run	Holds the status, stage, packet, trace and usage
`hypotheses`	Candidate causes for a run	
`evidence_links`	Citations from a hypothesis	
`investigation_similar_incidents`	Matches for a run	Score and reason
`tool_calls`	Each MCP call in a run	`seq` gives the `mcp:N` number
`feedback`	Votes, notes and verified causes	
`rca_drafts`	RCA versions	Linked to the incident and the run

The `evidence_packet` column keeps exactly what the model saw, so a run stays auditable after the source data changes. IDs are UUID strings; seeded IDs are derived from stable names, so a reset gives identical data.

Status values

- **Investigation:** `queued`, then `running`, then `completed` or `failed`.
- **Review of an incident:** `none`, `pending` after a completed run, and `verified` once a verified cause is recorded and the latest RCA is reviewed. It is computed by a query rather than stored.

Inspect the data safely

Use a copy, or point a temporary instance at its own file so you never touch shared data:

```
DB_PATH=/tmp/resolveai-test.db API_PORT=8799 dotnet run --project
src/ResolveAI.App
```

HTTP API

JSON uses snake_case, and errors look like `{ "error": "message" }`. An incident's `{id}` can be a UUID or a key such as `INC-1042`. Live requests send the access code in the `x-access-code` header.

Method and route	Purpose	Notable responses
`GET /api/config`	Mode, live availability, usage today	
`GET /api/systems`	Status and tools per system	
`GET /api/incidents`	List, with `status, severity, service, q, review, page, page_size`	
`GET /api/incidents/{id}`	Detail, runs and RCA versions	404 unknown
`GET /api/incidents/{id}/evidence`	Logs, signals, deployments, metrics, timeline, runbooks	404 unknown

Method and route	Purpose	Notable responses
`POST /api/incidents/{id}/investigations`	Start a run: `{question?, mode?}`	202; 400; 403; 429
`GET /api/investigations/{id}`	Status, hypotheses, trace, tool calls	404 unknown
`POST /api/investigations/{id}/feedback`	Vote, note or verified cause	201
`POST /api/incidents/{id}/rca-drafts`	Generate an RCA: `{investigation_id}`	201; 400 wrong incident; 409 not completed; 502 model error
`GET`, `PATCH /api/rca-drafts/{id}`	Read or edit a draft	
`GET /api/rca-drafts/{id}/export.md`	Markdown download	
`GET /api/knowledge/{id}`	Runbook or historical RCA	
`POST /api/demo/reset`	Reseed the demo	403 not localhost and no code; 404 demo mode off; 409 run active

Model errors map to 502 with a readable message, and unexpected errors return a generic 500.

8. Configuration

Configuration has two parts: environment variables for the app, and `mcp.config.json` for the systems it checks. Neither is required for a first run. Secrets belong only in `.env`, which is read at startup; real environment variables win over the file.

Environment variables

Variable	Default	Purpose
`ANTHROPIC_API_KEY`	empty	Claude API key. Empty means recorded mode
`LLM_MODE`	`recorded`	`live` makes Claude the default, if a key exists

Variable	Default	Purpose
`CLAUDE_MODEL`	`claude-opus-5-5`	Model for investigations and RCAs
`LIVE_ACCESS_CODE`	empty	Lets a run opt into live while the default stays recorded
`LIVE_DAILY_LIMIT`	`20`	Live model calls per UTC day
`DEMO_MODE`	`true`	Enables demo reset
`DEMO_RESET_ANY_IP`	`false`	Reset from any address. Docker only
`DB_PATH`	`data/resolveai.db`	SQLite file, relative to the repository root
`HOST`	`127.0.0.1`	Listen address
`API_PORT`	`8787`	Port
`MCP_ENABLED`	`true`	Start the MCP systems
`MCP_MAX_FOLLOWUP_CALLS`	`8`	Follow-up calls per run, 0 to 20
`RESOLVEAI_MCP_CONFIG`	empty	Config path. Otherwise `mcp.config.json`, then the example
`MSSQL_USER`, `MSSQL_PASSWORD`	empty	SQL login, for the `tedious` driver setting only

Docker Compose pins `HOST`, `API_PORT` and `DB_PATH`, so `.env` values for those are ignored in a container.

Turning on live mode

A live run needs an API key, and either `LLM_MODE=live` or a `LIVE_ACCESS_CODE` entered under Settings. Live mode has never been run for real on this project, and it costs money, so agree it with the project owner first. Treat the first run as a test.

mcp.config.json

Copy `mcp.config.example.json`. It is gitignored, so each machine has its own, and the app uses the example if there is none. Relative paths resolve against the file's folder. This shape comes from the example:

```
{
  "services": {
    "order-api": {
      "log_roots": ["demo-systems/logs/order-api"],
      "database": "ResolveAIDemo",
      "key_metrics": ["Order failures", "DB request p95"]
    }
  },
  "files": { "allowed_roots": ["demo-systems/logs"] },
  "deploys": { "feed": "demo-systems/deploys.json" },
  "metrics": { "series_dir": "demo-systems/metrics" },
  "sqlserver": {
    "enabled": false,
    "server": "localhost",
    "driver": "msnodesqlv8",
    "execute_as": "resolveai_reader",
    "trust_server_certificate": true
  }
}
```

The driver names are left over from the Node version, and are kept so that existing files stay valid. Both now use `Microsoft.Data.SqlClient`: `msnodesqlv8` means Windows sign-in, and `tedious` means a SQL login.

SQL Server setup (optional)

This turns on the real database checks and the INC-1046 blocking scenario.

```
sqlcmd -S localhost -E -C -i demo-systems/sqlserver/setup.sql
# then set "sqlserver": { "enabled": true } in mcp.config.json and restart
```

`setup.sql` needs an administrator once. It creates the `ResolveAIDemo` database and the `resolveai_reader` login. To remove it, run `teardown.sql`. To produce live blocking, follow the two-terminal steps at the top of `scenario-blocking.sql`.

9. Common engineering tasks

Most changes touch one small area, but `tests/ResolveAI.Tests/Golden` often needs a deliberate update too. Work on a branch, run the tests in a separate step, and only then commit.

Add a tool to an existing server

1. Add the tool in the right `src/ResolveAI.Mcp/*Server.cs`, with fixed, parameterised behaviour and read-only annotations through `ToolServer`.
2. Check every input a caller controls: paths, names and time windows.
3. Add attack tests in `tests/ResolveAI.Tests/McpTests.cs` for those inputs.
4. Update `Golden/tools-list.json`, which pins every tool's schema.
5. Add it to `Mcp/Baseline.cs` only if every investigation should run it. Otherwise Claude can choose it as a follow-up.

Add a new system

1. Add `<Name>Server.cs` in `src/ResolveAI.Mcp` that returns a `ToolServer`.
2. Register it in `ServerDefs` in `src/ResolveAI.Mcp/Program.cs`.
3. Add its colour to `SystemStyle` in `Components/Shared/Ui.cs`.
4. Add its baseline checks to `Mcp/Baseline.cs`.
5. Use a read-only credential, so that the system itself enforces the guarantee.
6. Add attack tests for everything a caller controls.

Never add a tool that takes free-form SQL or a command. That breaks the central safety rule.

Change a prompt or output schema

1. Edit `Llm/Prompts.cs` or `Llm/Schemas.cs`.
2. Change the prompt version, currently `inv-2026-10-28.1`, so that stored runs can be traced to their prompt.
3. Update `Golden/prompts.json` and explain the change in the review.
4. Recorded mode cannot show whether a prompt got better. Judging that needs a real, approved live run.

Change the database schema

1. Add a new numbered file, such as `004_description.sql`, in `Data/Migrations`. Never edit one that has shipped.
2. Migrations apply at startup in filename order, each in a transaction, and are recorded by filename.
3. Update the data access and DTOs, and the golden files if responses change.
4. Test on a copy of real data as well as a fresh database.

Add a redaction rule

1. Add the pattern to `Pipeline/Sanitizer.cs`.
2. Add a test beside the existing ones in `ApiGoldenTests.cs`, with a value that must be removed and a similar one that must stay.
3. Run the tests. Golden files show whether any existing packet changed.

Change styles

The generated `wwwroot/app.css` is committed. After adding or changing CSS classes, regenerate it:

```
npm install --prefix tools/css && node tools/css/build.mjs
```

Add or change a demo incident

Edit `Data/Seed.cs`. Seeded IDs derive from stable names, so renaming a record changes its ID and can break recorded citations. Then use **Reset demo data** on a temporary database, and update `Golden/seed.json`.

Update a golden file

Golden files hold what the original Node version returned, and Node has been removed. There is no tool to regenerate them, so edit the JSON by hand. Treat any golden difference as a bug unless you meant to change the behaviour, and say why in the review.

10. Testing and CI

Run `dotnet test tests/ResolveAI.Tests` before every commit. It needs no external services. Run the browser tests when you touch the UI.

Layer	Where	Proves	Does not prove
Golden files	`tests/ResolveAI.Tests/Golden`	25 routes, 7 runs, 37 tool results, schemas, seed and prompts match the original	That the original was right
API behaviour	`ApiGoldenTests.cs`	Citations, retries, failures, access code, limits, review states	Real model quality

Layer	Where	Proves	Does not prove
Attack tests	`McpTests.cs`	Path escapes, non-hex commits, unsafe login names	Unknown attack forms
Real processes	`HubAndClaudeTests.cs`	Servers start as children; the API key does not reach them	
Fake Claude API	`HubAndClaudeTests.cs`	Request contents, fallback, tool loop, error mapping	**That the real API accepts these requests**
Real SQL Server	`MSSQL_TEST=1`	Tools run as the reader; writes are refused	Production load
Browser	`tests/ResolveAI.E2E`	The full journey, inert HTML in logs, phone layout, refused cross-site connections	Browsers other than Chromium

Reading a failure

What fails	Usually means	Do this
A golden comparison	Behaviour changed, perhaps by accident	Read the difference. Fix the code, or update the file if it was intended
A test mentioning `MSSQL_TEST`	SQL Server is stopped or `setup.sql` was not run	Start it and run setup, or omit the variable
A browser test cannot launch	Chromium is not installed	Run the `playwright.ps1 install` command from section 2
Build error: file in use	A running app has locked the output	Stop the app and build again
A timing failure only in parallel runs	A startup race	Investigate; do not just retry. A race like this caused the MCP version pin

Browser tests use a temporary database and a free port. They must stay that way, because demo reset wipes data.

CI

`.github/workflows/ci.yml` runs on pushes to `main` and on pull requests. One job builds and runs all tests. The other builds the Docker image, starts it and checks the health of the key routes. **It has not yet run on GitHub**, so expect to debug the first run. The Linux unit tests and Docker checks were run locally; the browser tests have only run on Windows.

11. Operations and troubleshooting

Start every diagnosis with `GET /api/config` and the Systems page. Together they show the mode and each system's status, including the reason a system is not working.

Routine tasks

Task	Local	Docker
Start	`dotnet run --project src/ResolveAI.App`	`docker compose up -d --build`
Logs	Console output	`docker compose logs -f`
Stop, keeping data	Stop the process	`docker compose stop`
Back up	Copy `data/` while stopped	`docker compose cp resolveai:/app/data ./backup`
Upgrade	Pull and run again; migrations apply at startup	`docker compose up -d --build`
Reset demo data	Settings page	Settings page, or `docker compose down -v`

Reset erases all investigations, feedback and RCA drafts, and refuses while a run is active. Back up first if anyone's work matters.

Restart behaviour

- Runs in progress are marked failed at startup. Run them again.
- MCP servers start again with the app, and a crashed server is restarted once.

- Blazor pages reconnect, or may need a refresh.

Symptoms and causes

Symptom	Likely cause	Fix
A system shows *Not working*	The server failed to start or lost its connection	Read the detail on the Systems page
SQL Server shows as off	Off by default	Run `setup.sql` and set `enabled` in `mcp.config.json`
Database checks fail	SQL Server is stopped, or the reader login is missing	Start the service, or run `setup.sql`
Live option not offered	Needs both a key and an access code, or `LLM_MODE=live`	Check `/api/config`, then `.env`
HTTP 403 on a live run	Wrong or missing access code	Enter it under Settings
HTTP 429 on a live run	Daily limit used	Wait for the next UTC day, or use recorded mode
Access code forgotten after a Docker rebuild	Its encryption keys live inside the container	Enter it again
Blank page in a published build	Content root or client script missing	Publish with source present, so restore sees `.razor` files
Port 8787 in use	Another instance is running	Stop it, or set `API_PORT`
A run stays on one stage	A slow system or model	Wait for the time limit; the trace shows each stage's duration
Findings look wrong	Could be an analysis or an evidence problem	Compare the System checks tab, the trace and the stored packet
Findings look too good	Recorded mode is on	Check the LLM badge at the top right

Debugging a bad finding

1. Check the badge: is this recorded or live?
2. Open each cited evidence item and read the source record.
3. Check the System checks tab for failed calls.

4. Open **How this was produced** to see counts and durations per stage.
5. Read the validation issues. Removed citations and flagged numbers appear there.
6. Read the `evidence_packet` column if you need exactly what the model saw.

12. Security rules every engineer must keep

Treat everything from the browser, the model and the monitored systems as untrusted. The rules below are what keep that safe, so a change that weakens one needs an explicit decision.

Rules

1. **No write tools.** Never add a tool that changes a monitored system.
2. **No free-form queries or commands.** Tools take typed parameters. A database name is the only value placed into SQL, after an allow-list check and bracket quoting.
3. **Never bypass redaction.** Anything that reaches a model goes through the packet.
4. **Never bypass validation.** Citations, unsafe checks and numbers are checked on the server, whatever the model returns.
5. **Enforce limits in code, never in a prompt.** The follow-up call limit and the daily limit are enforced by ResolveAI.
6. **Keep rules in `AppService`.** A check added only to a route or a page leaves the other door open.
7. **Keep the child environment minimal.** MCP servers do not inherit the app's environment, so they never see the API key or access code.
8. **Keep evidence as data.** Prompts say evidence is not instructions. Do not render logs as raw markup in the UI.
9. **Keep secrets out of the repository.** They belong in `.env`, which is not committed. The seeded `sk_live_51FAKE…` token is fake and is there to test redaction.
10. **Bind to localhost.** The app has no sign-in. Do not expose it to a network as it stands.

How each system stays read-only

System	How
SQL Server	Each statement runs as `resolveai_reader`, with only read grants, and always reverts. The server refuses writes even on an admin connection

System	How
Files	The real path of every segment must be inside an allowed folder. `..`, symlinks, junctions, device paths and alternate streams are refused
Git	No shell, fixed arguments, repositories only from config, hex-only commit IDs
Metrics	Service names must be plain names, so they cannot become paths
All	The hub ignores any tool not marked read-only

Known security gaps

These exist today. Do not assume they are handled.

- **No user accounts or roles.** Anyone reaching the port can read everything and run recorded investigations.
- **No audit log** of who did what.
- **No cross-site protection on the HTTP API.** A web page you visit could send a request to localhost, including `POST /api/demo/reset`, which erases investigations. Set `DEMO_MODE=false` where that matters.
- **No DNS-rebinding protection.** `AllowedHosts` is `*`, and the Blazor origin check compares against the Host header.
- **Pattern-based redaction** can miss a secret. Think about this before sending real data to a cloud model.

13. Known gaps, traps and history

The biggest unknown is whether live analysis is actually useful, because Claude has never run against the real API on this project. Everything a demo shows comes from recorded mode.

Not proven

Item	Status
Live Claude	Never run for real, in Node or .NET. No API key is configured
Analysis accuracy	No evaluation set exists

Item	Status
GitHub CI	Written but never run on GitHub
Real data	Only SQL Server checks can be real today, plus files and git if pointed at real paths

Limitations

- **Point-in-time checks:** `db_blocking`, `db_health` and `db_long_queries` show the state now, not during the incident. They can mislead for an incident that has ended.
- **Keyword search only:** similar incidents phrased differently can be missed.
- **Single process:** a restart fails runs in progress, and SQLite limits it to one server.
- **Stale guide:** `docs/ResolveAI-Simple-Guide.pdf` still gives the old `npm` commands.
- **Stale text:** the Settings page says the recorded analysis covers INC-1042; it covers three incidents.

Traps that have caught people

Trap	What happened	Lesson
`.gitignore` hid source code	`data/` also matched `src/ResolveAI.App/Data` on Windows, so the data layer was not committed. Found by testing a fresh clone	Test a fresh clone before calling a change finished
MCP version probe race	Servers failed at random in about 40% of parallel runs, so the protocol version is pinned to 2025-11-25	Do not remove the pin without a reason and testing
Publishing the MCP executable	It needs the ASP.NET Core framework reference, or all servers crash at launch	Test the published app, not only `dotnet run`
Docker restore order	Restore must see `.razor` files, or `blazor.web.js` goes missing and the page is blank	Test the built image
Tests wiping data	Demo reset erases data, so browser tests must use a temporary database	Never point tests at `data/resolveai.db`

Trap	What happened	Lesson
Running a second copy	The default database is shared	Set `DB_PATH` and `API_PORT` for any extra instance
Git Bash on Windows	It rewrites `/app/...` paths in `docker exec`	Be aware when a path looks wrong

How it got here

The project began as a Node API with a React UI and Node MCP servers. It was ported to .NET 10 on `feature/dotnet-port`, and every route, investigation, prompt and tool result was recorded first as golden files. Those files are why the tests refer to Node. `main` still holds the Node version; merging is the project owner's decision.

The repository has two remotes: `origin` (the public `resolveai`) and `dotnet` (private `resolveai-dotnet`, which the branch tracks). Push to the right one.

Where it is going

The plan is in Part B of the architecture document. Nothing there is built.

1. Prove live Claude, with an evaluation set. This is a real decision point.
2. Windows sign-in with AD groups mapped to Viewer, Investigator, Reviewer and Admin.
3. Move from SQLite to SQL Server.
4. Host and operate it on Windows Server.
5. Bring in real incidents from one incident tool.
6. Add further evidence sources, one at a time.

14. Glossary and KT checklist

Glossary

Term	Meaning
Baseline check	A fixed read-only check run for every investigation, before any model is involved
Blazor Server	A .NET UI framework where page logic runs on the server, over a live connection
Circuit	Blazor's live connection between one browser page and the server

Term	Meaning
Evidence packet	The bounded, redacted set of records the model may see and cite
Follow-up check	An extra tool call chosen by Claude (live), or replayed (recorded)
FTS5	SQLite's full-text search engine, which ranks results with bm25
Golden file	A recorded correct output that tests compare against
Hypothesis	A candidate root cause, with citations, counter-evidence, unknowns and checks
MCP	Model Context Protocol: a standard way of offering tools to a model
RCA	Root cause analysis: the post-incident report
Recorded mode	Runs the pipeline with written or rule-based analysis and no model
Speculative	A hypothesis with no valid citation, forced to low confidence
stdio	Communication over a process's standard input and output

KT checklist

The handover is complete when the new engineer can tick every item without help.

Run it

- [ ] Cloned the repository and checked out `feature/dotnet-port`
- [ ] Ran the app, and completed an investigation and an RCA on INC-1042
- [ ] Ran `tests/ResolveAI.Tests` and `tests/ResolveAI.E2E`
- [ ] Ran the app in Docker
- [ ] Started a second instance with its own `DB_PATH` and `API_PORT`
- [ ] Optional: set up SQL Server and investigated INC-1046 during the blocking scenario

Explain it

- [ ] Can draw the processes from memory
- [ ] Can name the seven stages, and which depend on the provider

- [ ] Can explain the difference between observed facts and inferred analysis
- [ ] Can explain why recorded mode does not prove analysis quality
- [ ] Can explain how read-only is enforced for each system
- [ ] Can list the known security gaps

Change it

- [ ] Traced a citation from a hypothesis back to its source record
- [ ] Added a practice tool on a scratch branch, with an attack test
- [ ] Caused and read a golden test failure
- [ ] Stopped a system and diagnosed it from the Systems page

Handover

- [ ] Has access to both remotes and knows which one to push to
- [ ] Knows who approves paid live runs and merges to `main`
- [ ] Has read Part B of the architecture document